Spatial Maps in Neuroscience. The neuroscience literature provides a foundational perspective on spatial representations through studies of spatial maps in the brain. Early work on place cells in the hippocampus of rats revealed that specific neurons encode an animal's position in a two-dimensional space [44, 45]. Subsequent research identified other spatially tuned cells—such as grid cells, which encode grid-like patterns; head-direction cells, which signal the direction of gaze; and boundary cells, which respond to environmental edges [46–48]. The specialization observed in biological systems suggests that generalist artificial systems, including LLMs, might similarly develop internal representations for spatial orientation and navigation.

3 Evaluating LLMs in a Spatial Orientation Task

To evaluate spatial orientation in an agentic context, we designed the Grid-World Spatial Orientation Task (GWSOT), as illustrated in Fig. 1. This is a simple variation of a classic reinforcement learning task [49, 50], where an agent and a goal are placed in an 5×5 grid. The two left-most panels of Fig. 1 depict example trials, where green arrows indicate a correct path leading to the goal, whereas red arrows show a path that result in failure by reaching a maximum number of steps.

At every step, the agent is provided with the current world state in a conversation-style prompt—where user messages describe the grid state and assistant messages record previous decisions. The LLM is required to respond with a JSON object containing a single key, "action", whose value is one of four commands: "UP", "DOWN", "LEFT", or "RIGHT". An action is defined as correct if it reduces the Manhattan distance between the agent and the goal. A trial ends when the agent reaches the goal or when it exceeds a maximum step limit.

We use different Spatial Information Representations (SIRs) to convey the current world state to the LLM. We categorized SIRs into three classes:

- Cartesian representations: The LLM receives explicit (x, y) coordinates for the agent and the goal, along with the grid size.
- Topographic representations: The two-dimensional structure of the grid is preserved by representing cells and objects using characters or words. This requires LLMs to be able to interpret the text layout as structure, which previous research has shown these models are able to achieve [51].
- Textual representations: The world state is described in prose-like language.

To ensure that our findings are attributable to the SIR class rather than to a specific formatting style, we implemented two variants for each class: JSON and Chess Notation for Cartesian, Symbol and Word Grid for Topographic, and Row and Column Description for Textual (see Fig. 1, right panels). Despite their differences, all six SIR types encode identical spatial information.

For additional details on the task configuration and system prompt see Appendix A.

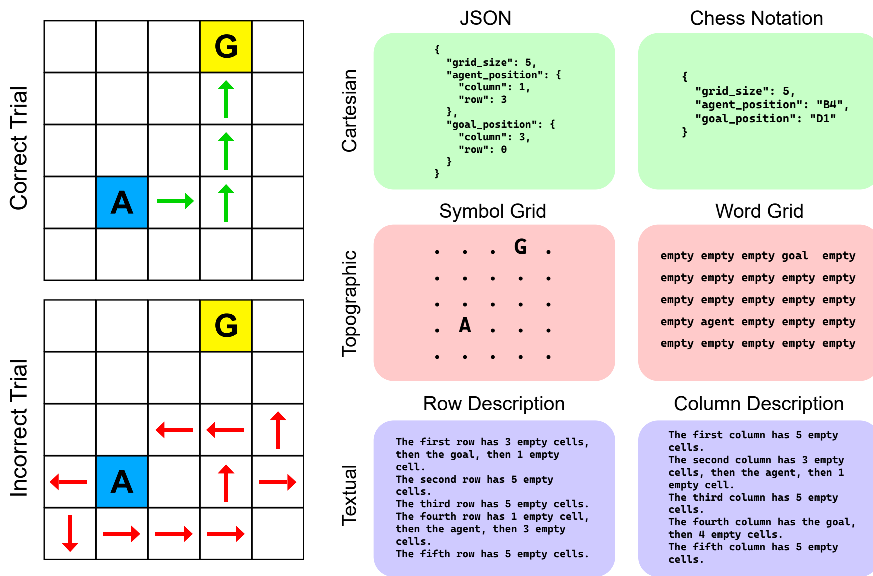

Fig. 1. The Grid-World Spatial Orientation Task (GWSOT). A Goal (G, yellow) is placed in a random position in a 5×5 grid (left-most panels). An Agent (A, blue) is placed in a semi-random location in the same grid, at least two steps away from the Goal. Green arrows (top panel) show a correct path which leads to the goal. Red arrows (bottom panel) shows an incorrect path which leads to task failure. The six panels on the right show examples of the three SIR classes and six SIR types used across this study.

4 Prompting and Scale as Factors in Performance

We evaluated the LLaMA 3 family of models—specifically the 3.2-1B, 3.2-3B, 3.1-8B, 3.2-11B, 3.1-70B, and 3.2-90B variants—in the 5×5 GWSOT. For each model and each of the six SIR types, we conducted 100 trials, resulting in a dataset of 3,600 trials. In addition, we ran 100 trials with a random policy agent that uniformly selected one of the four possible actions at each step, providing a baseline for comparison. A complete account of each experiment performed in this work, along with their features and number of trials per condition can be found in Appendix B.

Model performance was quantified using three metrics. First, we computed the success rate as the proportion of trials in which the agent reached the goal. Second, for successful trials, we measured path efficiency by dividing the minimum number of steps required to reach the goal (i.e., the initial Manhattan distance) by the actual number of steps taken by the agent; an efficiency of 1 indicates a perfect, direct path. Third, for unsuccessful trials, we calculated the final distance ratio by dividing the final Manhattan distance from the agent to the goal by their

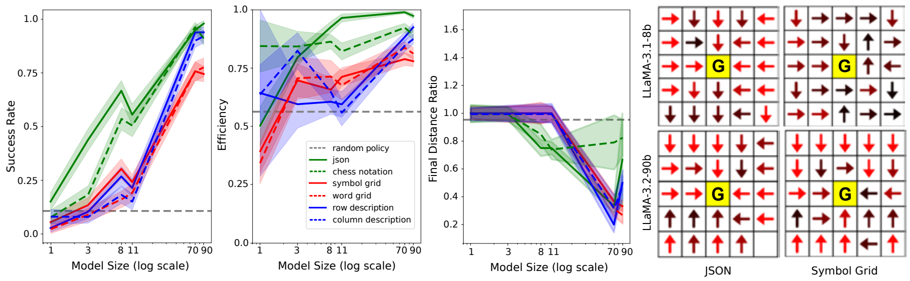

Fig. 2. Model performance improves with model size and is influenced by SIR type. The first three panels show performance metrics in the 5×5 GWSOT for Cartesian (green), Topographic (red) and Textual (blue) SIR classes. In all cases, the gray dashed line shows the performance of a random policy agent. Left-most panel shows the success rate of LLMs in this task as a function of model size. Second panel show the mean efficiency of LLMs in this task as a function of model size, only for trials where the goal was reached. Third panel shows the mean final distance ratio as a function of model size, only for trials where the goal was not reached. In all cases, the x axis (model size) is logarithmic and shadings represent standard error. The right-most four panels depict example policy maps for the LLaMA-3.1-8B and the LLaMA-3.2-90B models, for the JSON and Symbol Grid SIRs. Arrows represent the most common action chosen by the model in each relative position to the goal. Brighter red arrows represent more commonly taken actions, whereas darker arrows represent uncertain decisions.

initial Manhattan distance. A ratio of 1 implies no improvement, while lower ratios indicate that the agent moved closer to the goal despite not reaching it.

The first three panels of Figure 2 summarize how both model scale and SIR type influence these performance metrics. The left panel shows that success rates increase with model size for every SIR type (Linear Regression, $\beta = 0.008$, $p < .001$). The smallest models (1B and 3B) perform near chance level—around 10%—for all SIR types except for JSON, while the largest models (70B and 90B) always exceed a 74% success rate. Particularly, the 90B model receiving the JSON representation achieved an outstanding 98% success rate. Notably, Cartesian SIRs consistently outperformed Topographic and Textual SIRs across model sizes (Binomial GLM: success $\sim$ representation + model_size, $\beta_{\text{topographic}} = 1.385$, $p < .001$; $\beta_{\text{textual}} = 1.270$, $p < .001$), although the mid-sized models (8B and 11B) exhibited the most pronounced performance differences between SIRs. For example, the 8B model achieves a 66% success rate with the JSON SIR but only 30% with its best non-cartesian SIR (Symbol Grid). Importantly, this pattern holds regardless of formatting variations within each SIR class (compare same-color curves in the left panel of Fig. 2—both JSON and Chess Notation outperform Topographic and Textual SIRs).

A similar trend was observed in the efficiency metric, shown in the second panel of Fig. 2. This metric also improved with the scale of the model (Gaussian GLM: efficiency $\sim$ model_size, $\beta_{\text{model\_size}} = 0.001$, $p < .001$), and Cartesian SIRs consistently resulted in models taking more efficient paths to the goal during